

Lecture 12 (Transport 3)

Congestion Control, Part 1

Lecture 12, Spring 2026

Congestion Control: Why do we need it?

Lecture 12, Spring 2026

Congestion Control Principles

- **Why do we need it?**
- Why is it hard?
- Design Goals

Dynamic Adjustment

- Algorithm Sketch
- Reacting to Congestion (AIMD)
- Implementing Congestion Control with Event-Driven Updates

Recall: Congestion at Routers

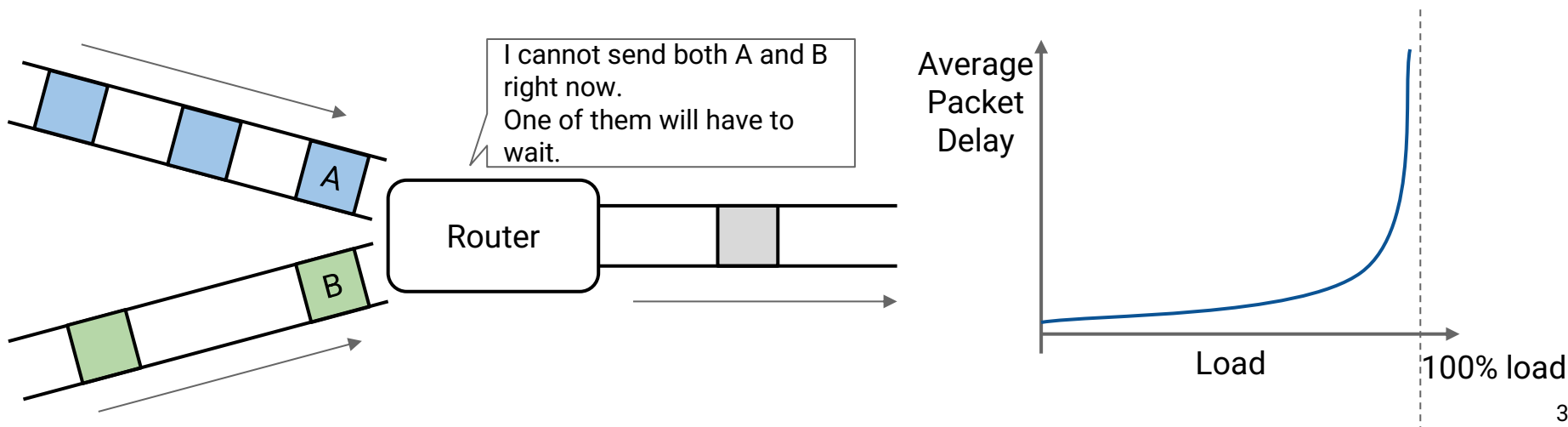
If two packets arrive at the same time, the router will send one, and buffer the other.

The router maintains a *queue* of packets that still need to be sent. If too many packets arrive close in time:

- The router cannot keep up, and gets congested.
- Packets can be delayed (stuck waiting in queue) or dropped (queue is full).

For a typical queuing system with bursty arrivals:

- Higher load = more packets sent = higher packet delay.
- Delay gets really bad, even before we reach 100% load.
- Trade-offs: Need to balance high link utilization and low delay.



In the 1980s, TCP did not implement congestion control.

- Sending rate was only limited by flow control (recipient buffer capacity).
- If packets are dropped, resend them over and over, at the same fast rate.

This led to a *congestion collapse*.

- 1986: Capacity of a major link dropped to 0.1% of its original capacity.
- Why? Network was flooded with thousands of copies of every packet.
- Michael Karels and Van Jacobson changed a few lines of TCP code in BSD (an early OS) to fix the problem.
 - The fix worked, and everybody quickly adopted it.
- Extensive research and improvements since the original patch.

Congestion Control: Why is it hard?

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Congestion Control Principles

- Why do we need it?
- **Why is it hard?**
- Design Goals

Dynamic Adjustment

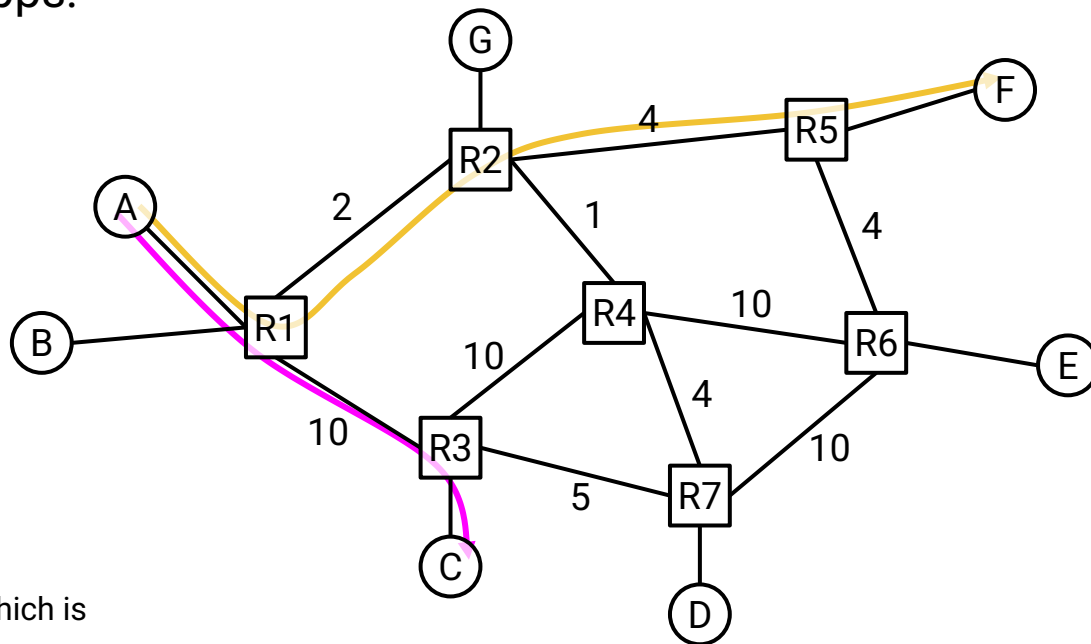
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Congestion Control Challenges: Depends on Destination

At what rate should A send traffic?

It depends on the destination.

- $A \rightarrow C$ can send at 10 Gbps.
- $A \rightarrow F$ can only send at 2 Gbps.



Numbers denote bandwidth in Gbps.

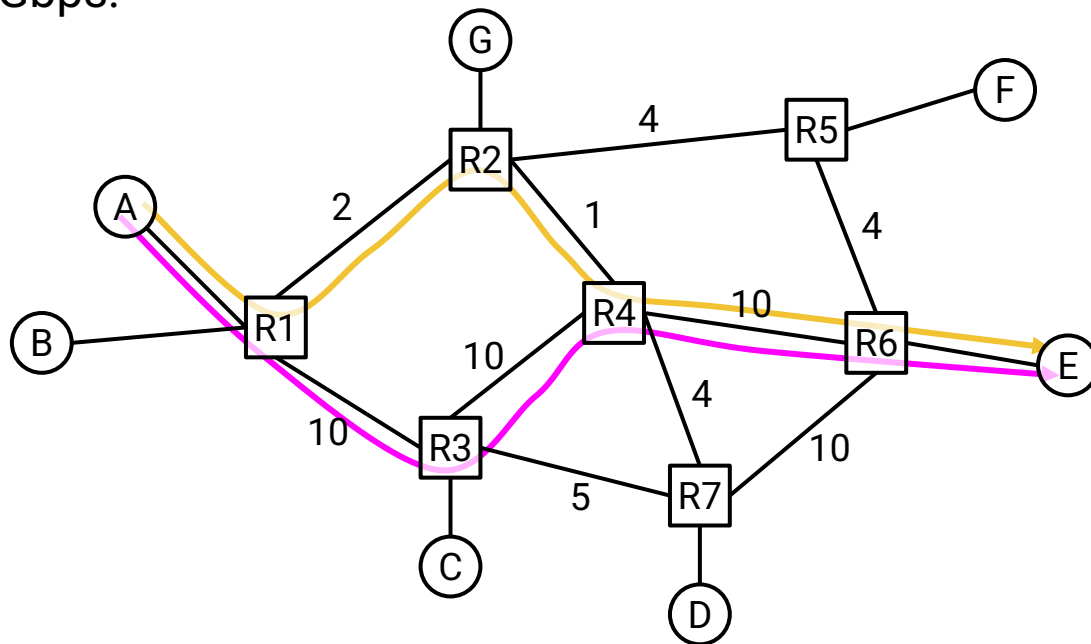
Ignore bandwidth of host-to-router LAN links, which is very high compared to router-to-router WAN links.

Congestion Control Challenges: Depends on Path

At what rate should $A \rightarrow E$ send traffic?

It depends on the path through the network.

- $A \rightarrow E$ via R3 can send at 10 Gbps.
- $A \rightarrow E$ via R2 can send at 1 Gbps.

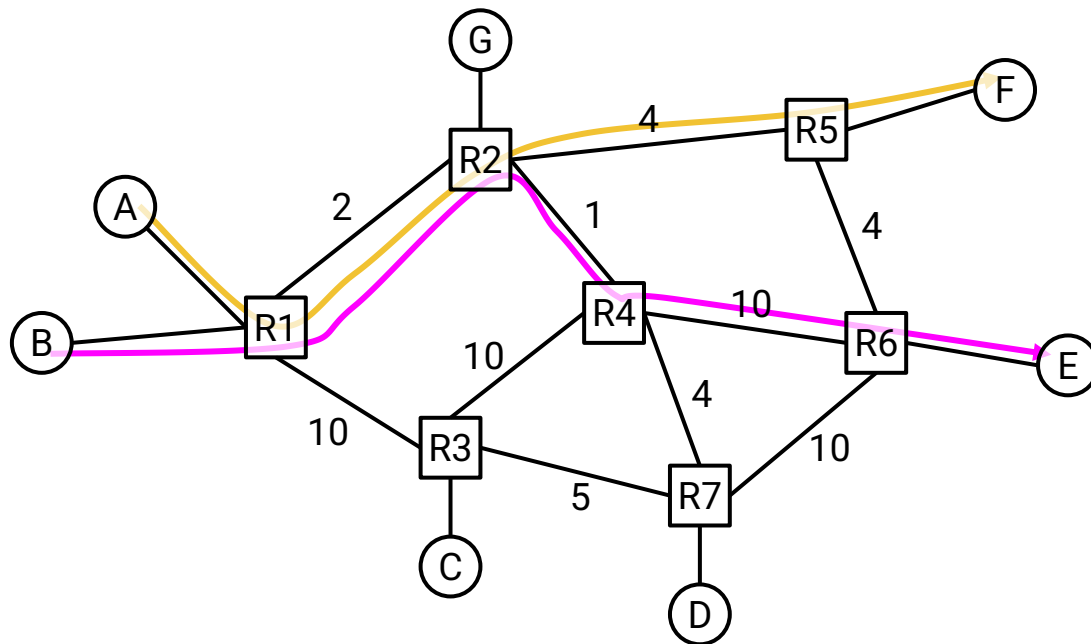


Congestion Control Challenges: Depends on Competing Flows

At what rate should $A \rightarrow F$ send traffic?

It depends on others using the network.

- $A \rightarrow F$ and $B \rightarrow E$ share a link.
- $B \rightarrow E$ can send at 1 Gbps.
- $A \rightarrow F$ can send at 1 Gbps.

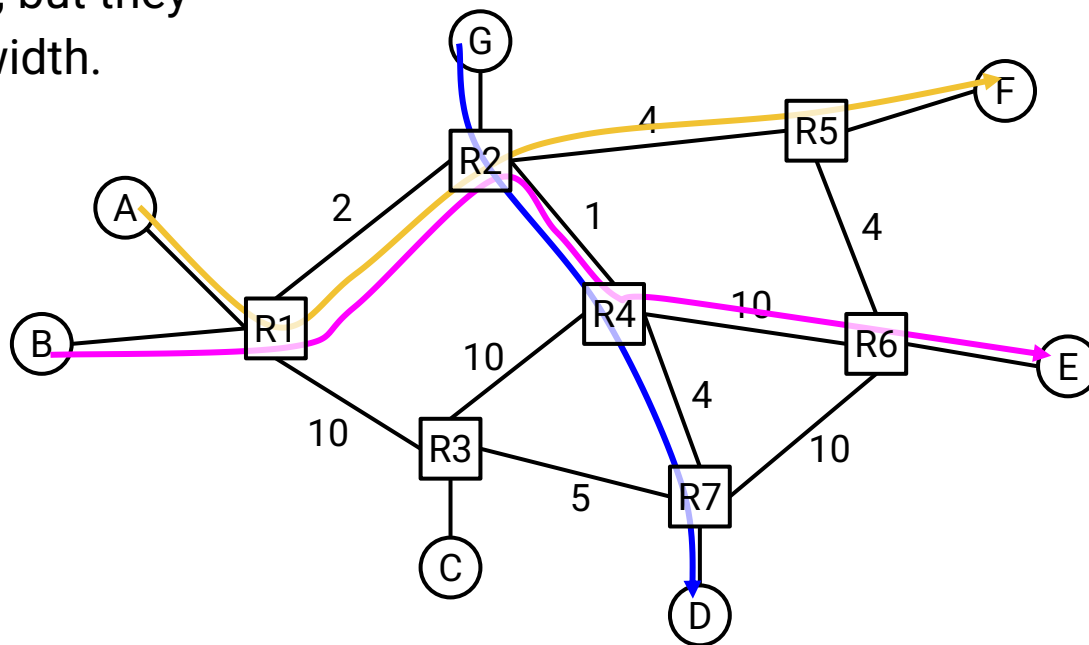


Congestion Control Challenges: Depends on Indirect Competition

What if $G \rightarrow D$ starts sending traffic?

- Maybe $B \rightarrow E$ should slow to 0.5 Gbps.
- Then $A \rightarrow F$ could increase to 1.5 Gbps.

$G \rightarrow D$ and $A \rightarrow F$ share no links, but they still affected each other's bandwidth.



The sender's optimal rate depends on:

- The destination.
- The path to the destination.
- Other connections sharing links along that path.
- Connections sharing links with those other connections.
- And so on...

Congestion control is a global problem.

- Connections in the network are highly interdependent.
- Solving it optimally requires a global view of the network.

Fundamentally, congestion control is a *resource allocation problem*.

- Bandwidth is a limited resource.
- Each connection demands a certain amount of bandwidth.
- We decide how much to allocate to each connection.

Resource allocation is a classic problem (e.g. CPU scheduling, memory allocation).

But, congestion control has additional challenges.

- A change in one connection can have a global impact.
- Network topology is constantly changing.
- Connections are constantly starting and ending.
- Everybody must solve it locally. No global view of network.

Design Goals

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Congestion Control Principles

- Why do we need it?
- Why is it hard?
- **Design Goals**

Dynamic Adjustment

- Algorithm Sketch
- Reacting to Congestion (AIMD)
- Implementing Congestion Control with Event-Driven Updates

Goals for a good congestion control algorithm:

1. Avoid congestion: Minimize packet delay and loss.
2. Efficiency: Maximize link utilization.
3. Fairness: Connections should share the available bandwidth.
4. Practical to implement: Scalable, decentralized, adaptive, etc.

Our ultimate goal is a good tradeoff between these metrics.

Reservations:

- Connections reserve bandwidth at the start, and release bandwidth at the end.
- Requires connections to know bandwidth ahead of time.
- Other problems: Hard to implement, inefficient bandwidth usage, etc.

Pricing/priorities:

- Analogy: Express toll lanes on the highway. Price changes depending on traffic.
- Prioritize traffic from users who pay more.
- Requires a payment model.

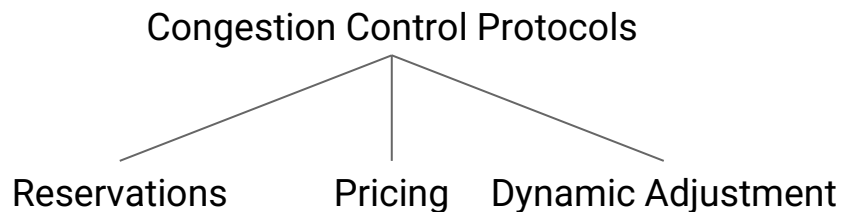
Dynamic adjustment:

- Learn the current congestion level, and adjust your rate accordingly.
- Requires good citizenship (no cheating).

3 possible approaches: reservations, pricing, and dynamic adjustment.

All the algorithms we'll see are based on dynamic adjustment.

- If we rebuilt the Internet from scratch, another approach might be feasible.
- But congestion control was invented as a patch on the existing Internet.
- Dynamic adjustment is the best fit for the existing Internet:
 - Doesn't require knowing bandwidth ahead of time.
 - Doesn't require a payment model.



Dynamic Adjustment: Algorithm Sketch

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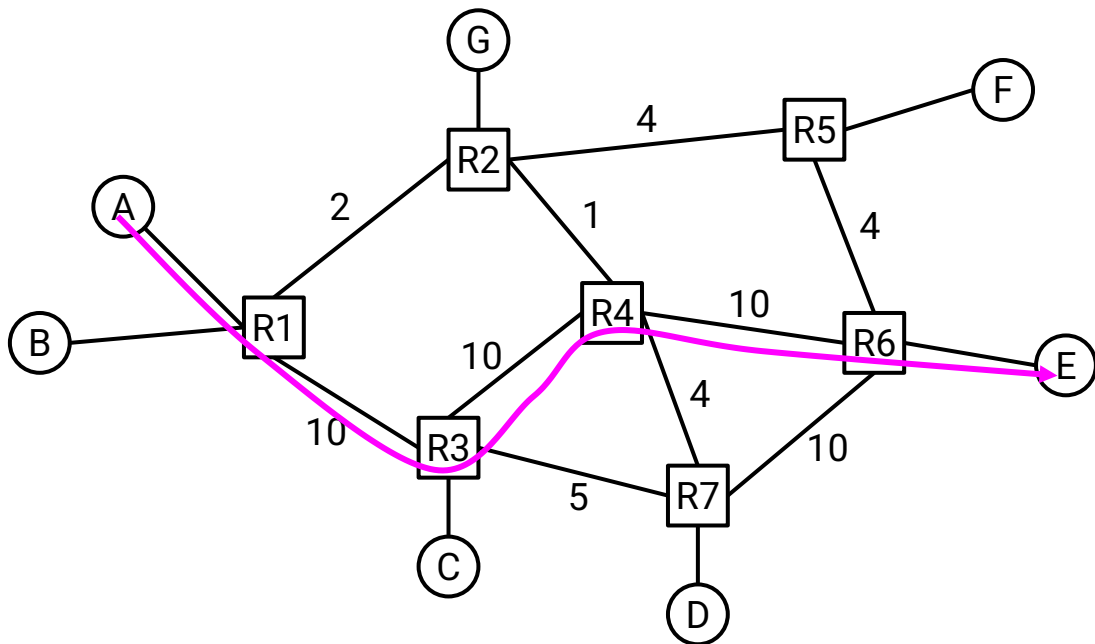
Congestion Control Principles

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Discovery: Host A discovers it can send data at 10 Gbps.

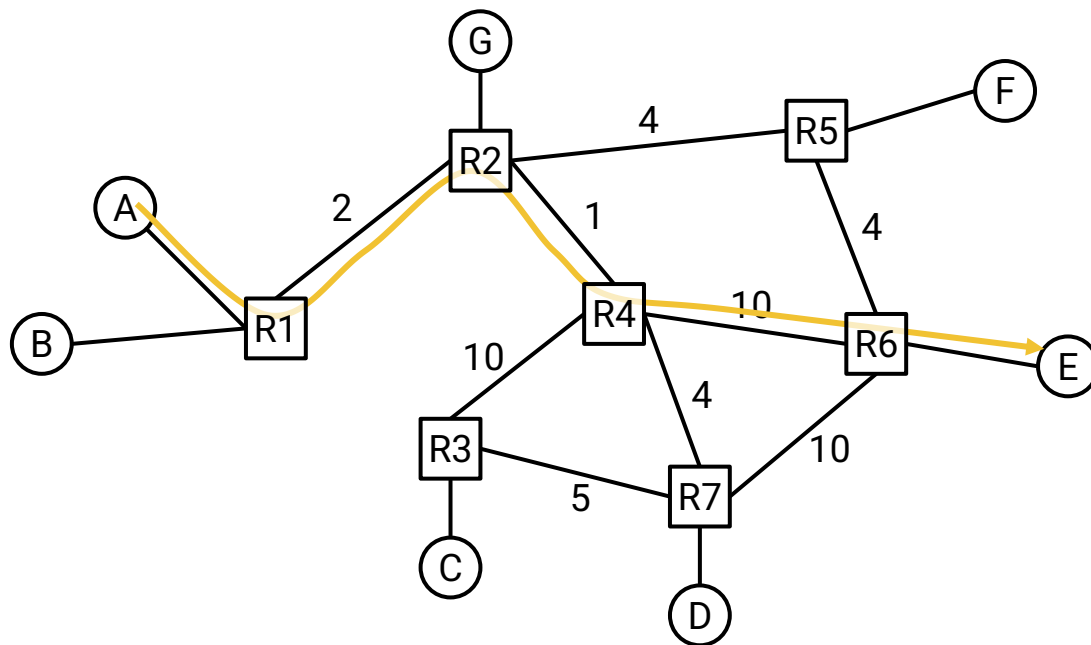


Dynamic Adjustment: Detection and Adjustment

Later, the $R1 \rightarrow R3$ link goes down. A's traffic is rerouted.

Detection: A notices that 10 Gbps is congesting the network.

Adjustment: A decides to slow down to 1 Gbps.

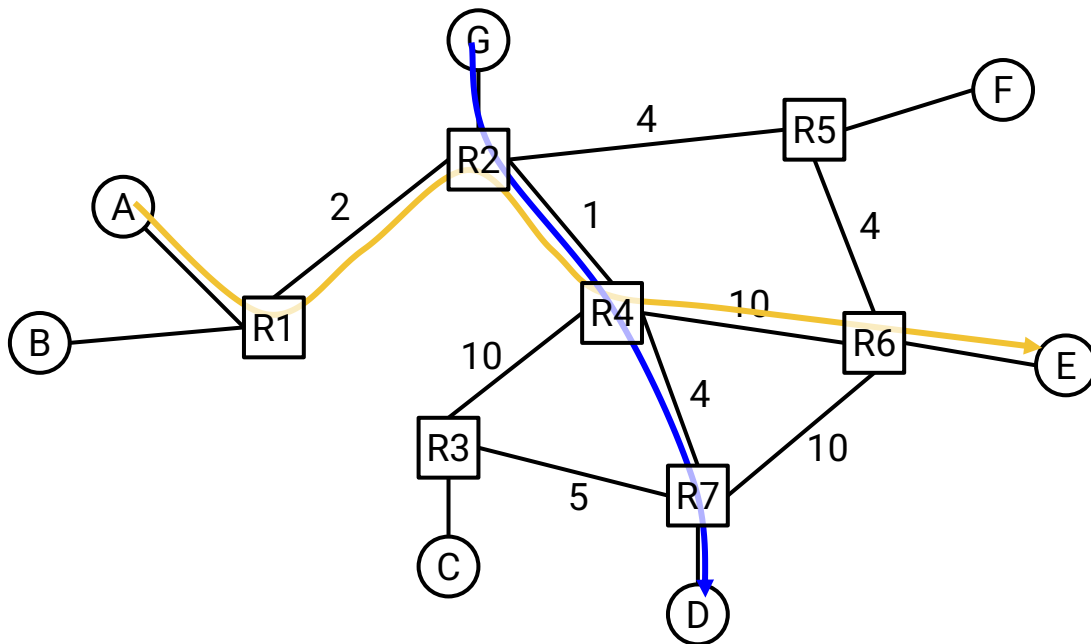


Dynamic Adjustment: Detection and Adjustment

Later, $G \rightarrow D$ starts sending traffic.

Detection: A notices that 1 Gbps is congesting the network.

Adjustment: A decides to slow down to 0.5 Gbps.



Types of Dynamic Adjustment Algorithms

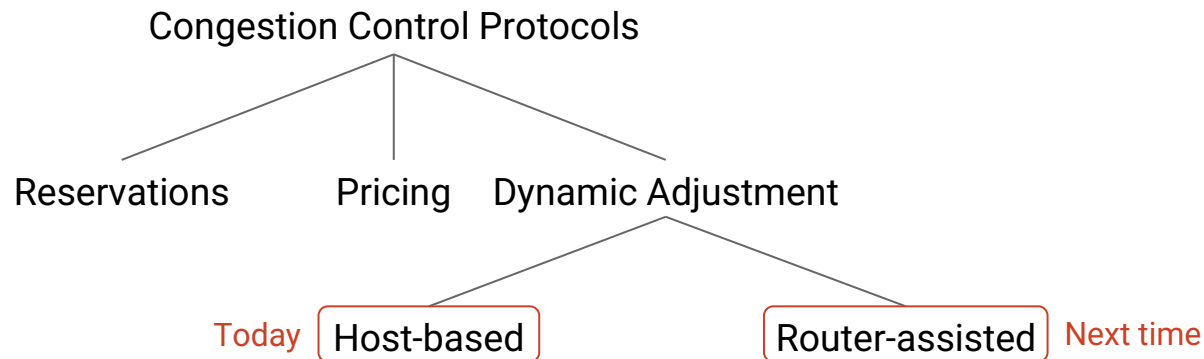
Solutions can be classified into 2 types, depending on where they're implemented:

Host-based congestion control: (*TCP uses this one*)

- Implemented at the end hosts. No special support from routers.
- Hosts adjust rate based on *implicit* feedback from routers.

Router-assisted congestion control:

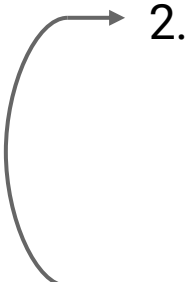
- Routers signal congestion back to end hosts.
- Hosts pick rate based on *explicit* feedback from routers.



Every host *independently* runs the same algorithm:

1. Pick an initial rate R . (*How do we pick the initial rate?*)

2. Try sending at rate R for some period of time.

- Did I experience congestion in this time period? (*How do we detect congestion?*)
 - If no, increase R .
 - If yes, reduce R . (*How much do we increase/decrease by?*)
 - Repeat.
- 

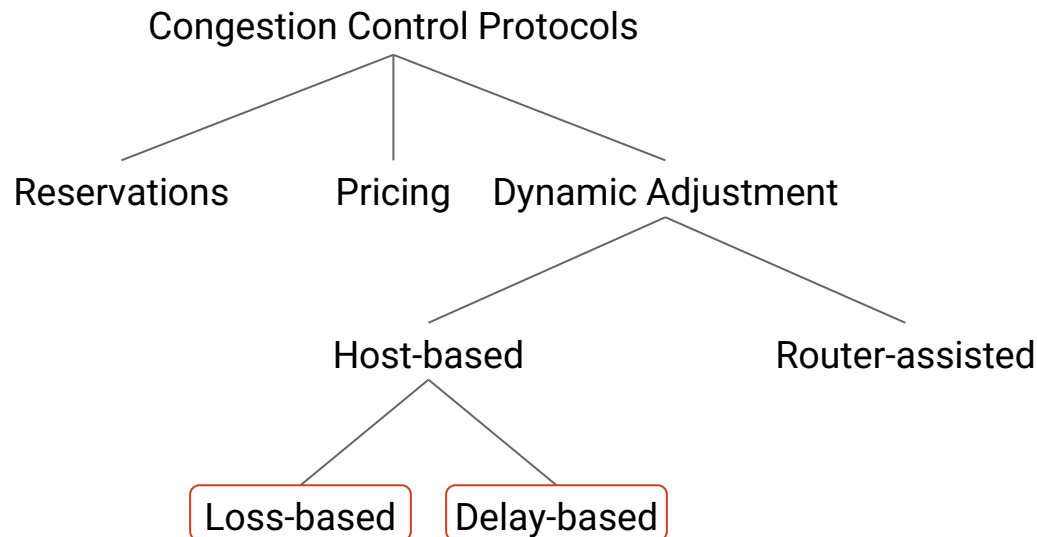
Components of a solution:

- Discovering an initial rate.
- Detecting congestion.
- Reacting to congestion (or lack thereof) by increasing/decreasing rate.

Detecting Congestion

There are 2 ways for the host to detect congestion.

Remember: The host can't see the whole network.



Loss-based detection: If we lose a packet, declare congestion. (This is what TCP uses.)

Pros:

- Unambiguous signal: Every packet is either lost (resent), or not lost.
- TCP already detects loss for reliability.

Cons:

- Not all loss is caused by congestion. (e.g. checksum error).
- A delayed packet could be mistakenly declared lost.
- When we detect loss, the network is already congested (queues are full).

Delay-based detection: If packets are delayed, declare congestion.

- Challenge: Packet delay length varies with queue size and other traffic.
- Long considered tricky to get right.

Discovering an Initial Rate: Slow Start

Goal for discovering initial rate: Estimate available bandwidth.

Start slow for safety.

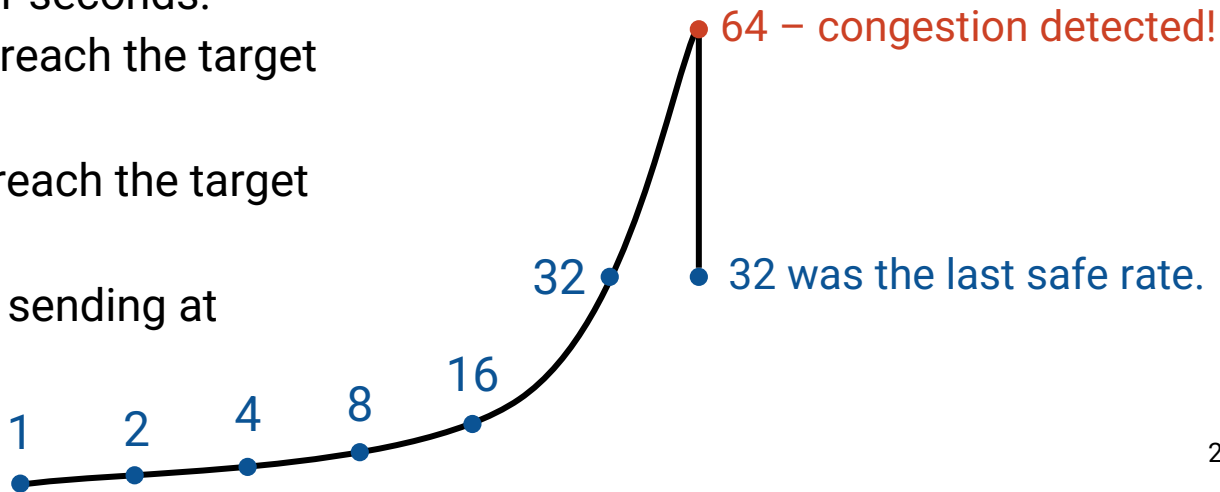
- Starting too fast would cause congestion.

Ramp up quickly for efficiency.

- Example of inefficiency: Suppose we added 0.5 Mbps every 0.1 seconds.
- If 1 Mbps available, we'll reach the target in 0.2 seconds.
- If 1 Gbps available, we'll reach the target in 200 seconds.
 - Lots of time wasted sending at suboptimal rates.

Slow start is an algorithm for discovering the initial rate:

- Start with a small rate (could be much less than actual bandwidth).
- Increase exponentially (e.g. double rate each time) until we detect loss.
- A safe rate is the most recent rate before detecting loss.
 - $0.5 \times$ rate when loss occurred.



Reacting to Congestion

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- Why do we need it?
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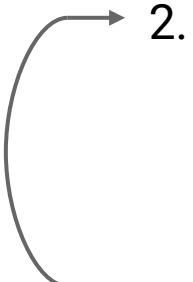
Dynamic Adjustment

- Algorithm Sketch
- **Reacting to Congestion (AIMD)**
- Implementing Congestion Control with Event-Driven Updates

Every host *independently* runs the same algorithm:

1. Pick an initial rate R . (*Slow start.*)

2. Try sending at rate R for some period of time.

- Did I experience congestion in this time period? (*Detect packet loss.*)
 - If no, increase R .
 - If yes, reduce R . (*How much do we increase/decrease by?*)
 - Repeat.
- 

Rate adjustment: How much do we increase/decrease by?

- Critical part of congestion control design.
- Determines how quickly a host adapts to changes in available bandwidth.
- Determines how effectively bandwidth is consumed.
- Determines how bandwidth is shared (fairness).

Goals for rate adjustment:

- Efficiency: High utilization of link bandwidth.
- Fairness: Each flow gets an equal share of bandwidth.

At a high level, we can either adjust quickly or slowly.

- Fast: **Multiplicative** changes.
 - Increase by doubling: $R \rightarrow 2R$
 - Decrease by halving: $R \rightarrow R/2$
- Slow: **Additive** changes.
 - Increase by adding: $R \rightarrow R + 1$
 - Decrease by subtracting: $R \rightarrow R - 1$

We can combine these rates in four ways:

- AIAD: Additive Increase, Additive Decrease.
- AIMD: Additive Increase, Multiplicative Decrease.
- MIAD: Multiplicative Increase, Additive Decrease.
- MIMD: Multiplicative Increase, Multiplicative Decrease.

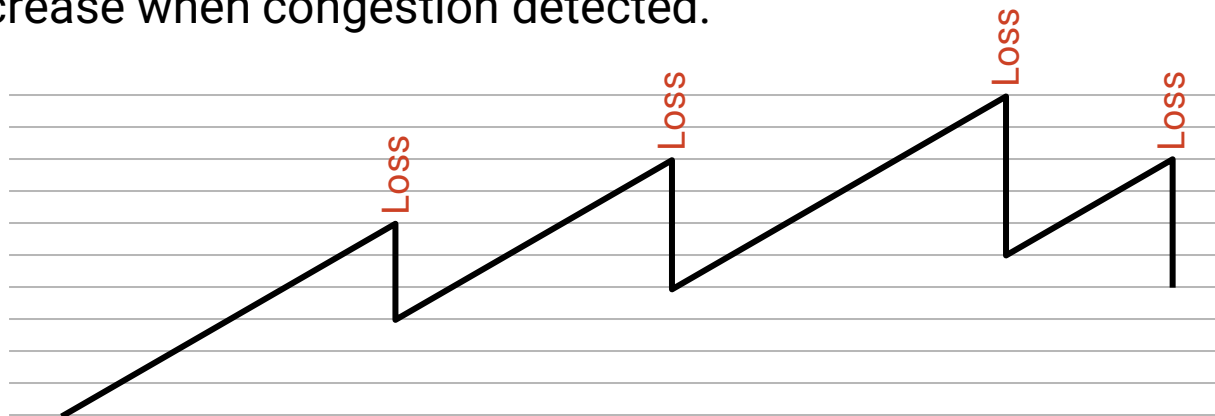
AIMD (Additive Increase, Multiplicative Decrease) is the best option.

Intuition: Sending too much is worse than sending too little.

- Sending too much: Congestion, packets dropped and retransmitted.
- Sending too little: Somewhat lower throughput.

General approach:

- Gentle increase when uncongested. (exploration)
- Rapid decrease when congestion detected.



Of the four options, AIMD is the only one that ensures fairness.

To see why, let's build a simple model:

- Two flows using a single link of fixed capacity C .
- Both flows are *independently* running *identical* dynamic adjustment algorithms.
- Flows are sending at rates X and Y .
 - If $X + Y > C$, the network is congested.
 - If $X + Y < C$, the network is underloaded.
- Goals:
 - Full link utilization: $X + Y = C$.
 - Fair sharing: $X = Y$.

Rate Adjustment Graph

Assume $C = 1$.

Efficiency line: $X + Y = 1$

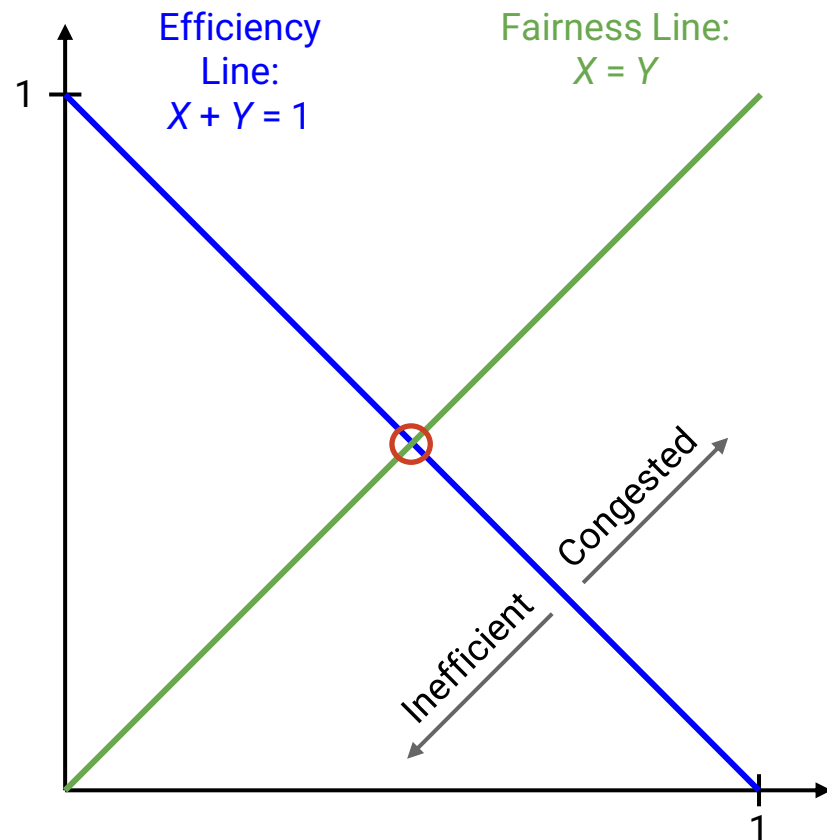
Congestion: $X + Y > 1$: above the efficiency line.

Unused capacity: $X + Y < 1$: below the efficiency line.

Fairness line: $X = Y$

The lines intersect at $(0.5, 0.5)$.

This allocation is both fair and efficient.



Rate Adjustment Graph

$(0.2, 0.5)$ is inefficient (below blue line).

Only 0.7 bandwidth used.

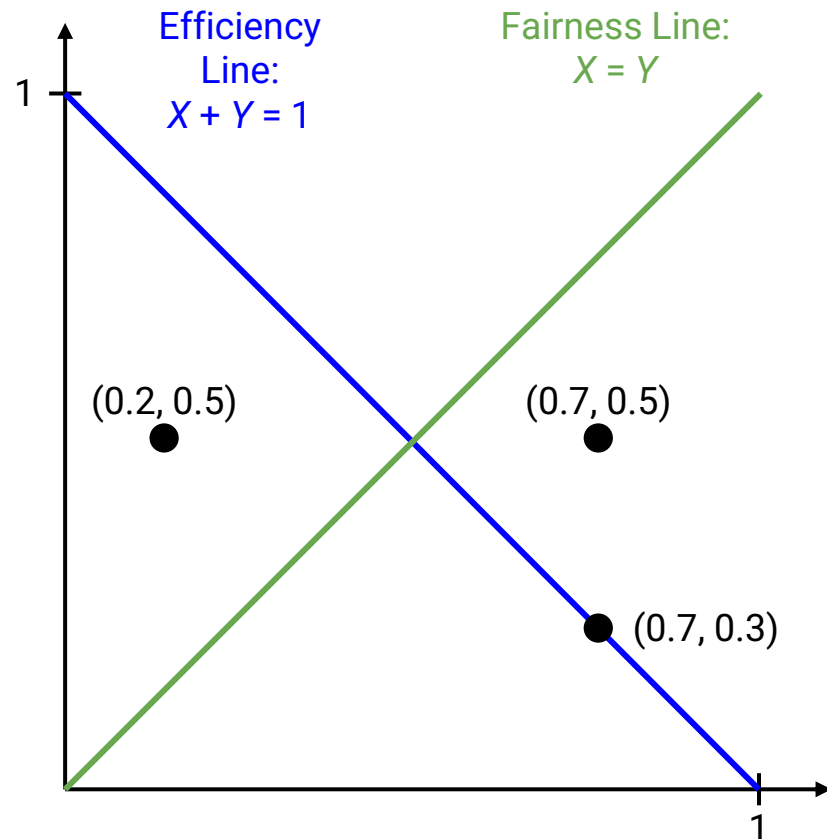
$(0.7, 0.5)$ is inefficient (above blue line).

1.2 bandwidth used (capacity is $C = 1$).

$(0.7, 0.3)$ is efficient, but unfair.

On blue line, but not on green line.

1 bandwidth used, but, $0.7 \neq 0.3$.



Additive Adjustments on Graph

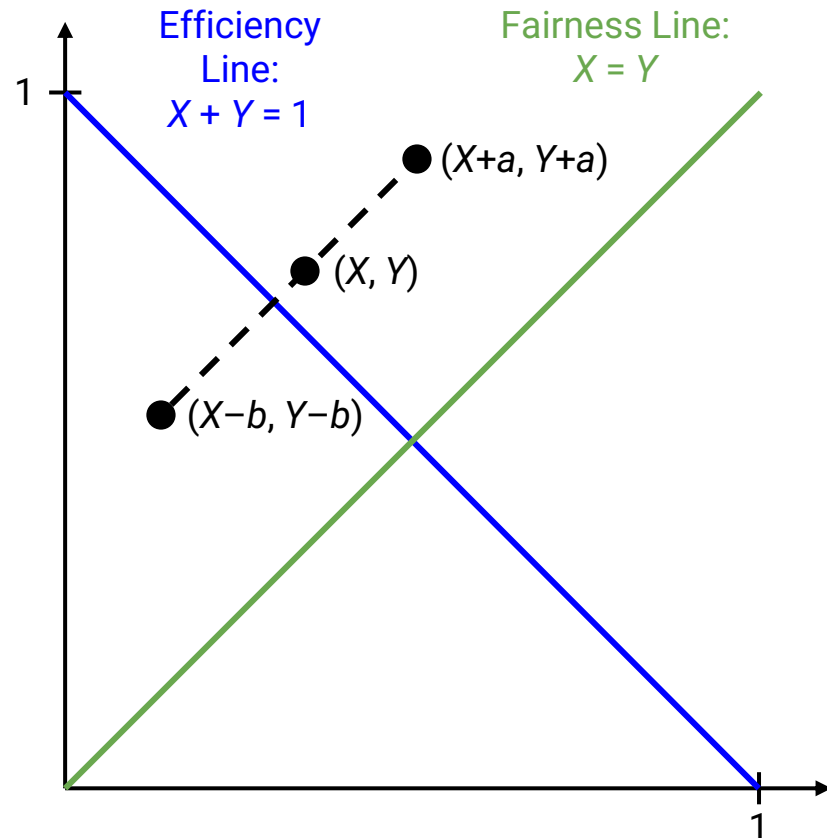
Recall: Both flows *independently* running *identical* algorithms.

Suppose current allocation is (X, Y) .

If both increase additively: $(X + a, Y + a)$.

If both decrease additively: $(X - b, Y - b)$.

Notice: Allocations always stay on a line with slope = 1.



Multiplicative Adjustments on Graph

Recall: Both flows *independently* running *identical* algorithms.

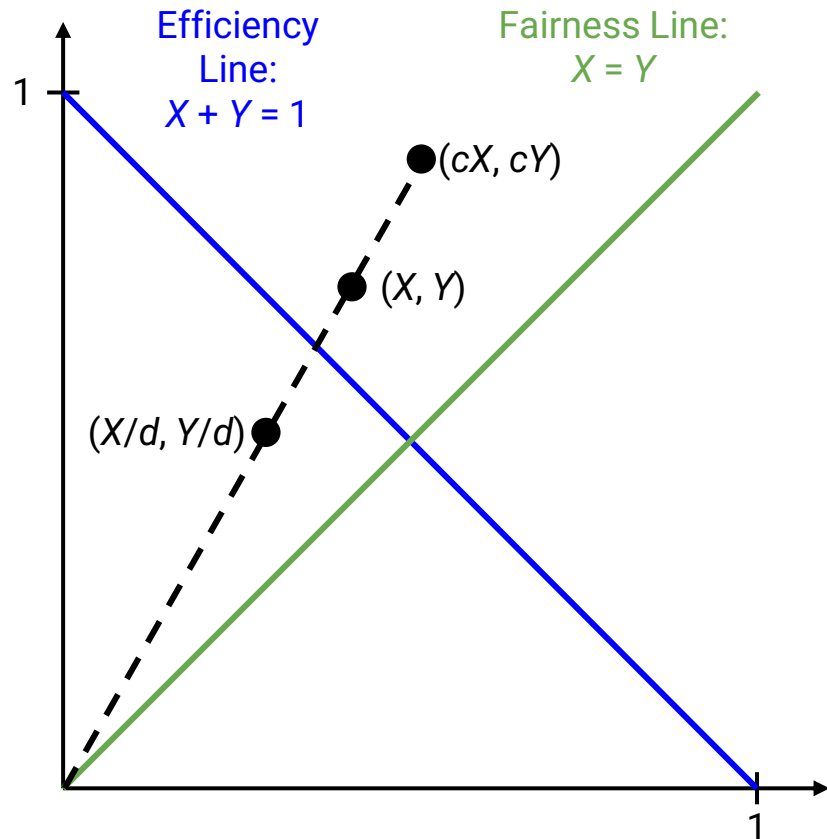
Suppose current allocation is (X, Y) .

If both increase multiplicatively: (cX, cY) .

If both decrease multiplicatively: $(X/d, Y/d)$.

Notice: Allocations always stay on a line with slope $= Y/X$, going through the origin.

$$\frac{\text{Rise}}{\text{Run}} = \frac{cY - Y}{cX - X} = \frac{Y(c - 1)}{X(c - 1)} = \frac{Y}{X}$$



AIAD (Additive Increase, Additive Decrease) Dynamics

Increase: +1

Decrease: -2

$C = 5$

Start:	$X = 1$	$Y = 3$	sum = 4, no congestion	$Y - X = 2$
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First iteration:	$X = 2$	$Y = 4$	sum = 6, congested	$Y - X = 2$
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Second iteration:	$X = 0$	$Y = 2$	sum = 2, no congestion	$Y - X = 2$
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Third iteration:	$X = 1$	$Y = 3$	back to where we started!	$Y - X = 2$
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Notice: The *difference* in allocations stays the same!

AIAD maintains the same unfairness across iterations.

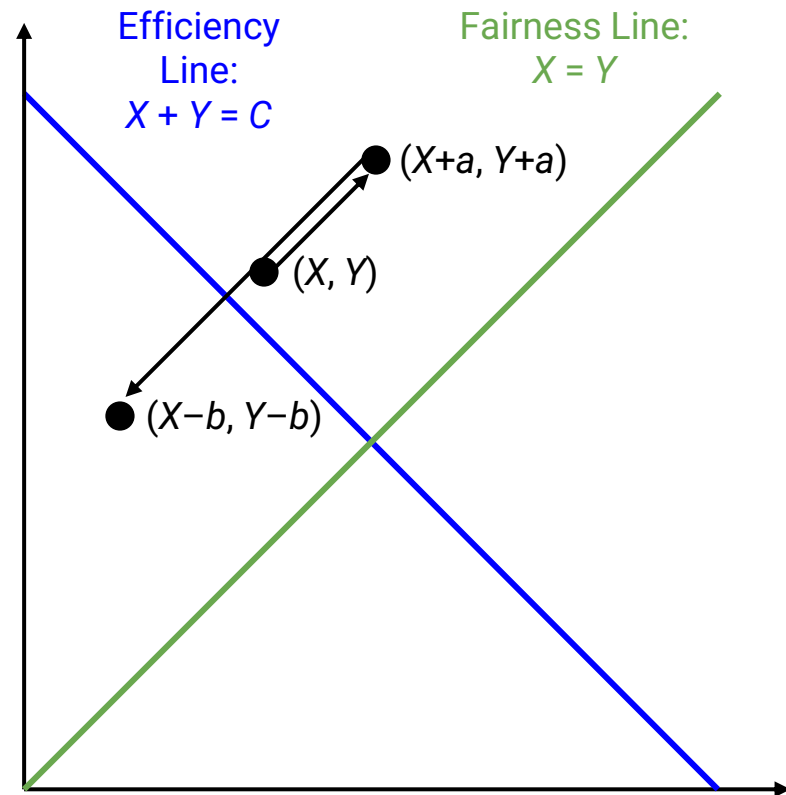
AIAD (Additive Increase, Additive Decrease) Adjustments on Graph

Increase: $+a$

Decrease: $-b$

We keep moving along the line with slope 1.

AIAD does not converge to fairness.



MIMD (Multiplicative Increase, Multiplicative Decrease) Dynamics

Increase: $\times 2$

Decrease: $\div 4$

$C = 5$

Start:	$X = 0.5$	$Y = 1$	sum = 1.5, no congestion	$Y/X = 2$
First iteration:	$X = 1$	$Y = 2$	sum = 3, no congestion	$Y/X = 2$
Second iteration:	$X = 2$	$Y = 4$	sum = 6, congestion	$Y/X = 2$
Third iteration:	$X = 0.5$	$Y = 1$	back to where we started!	$Y/X = 2$

Notice: The *ratio* of allocations stays the same!

MIMD maintains the same unfairness across iterations.

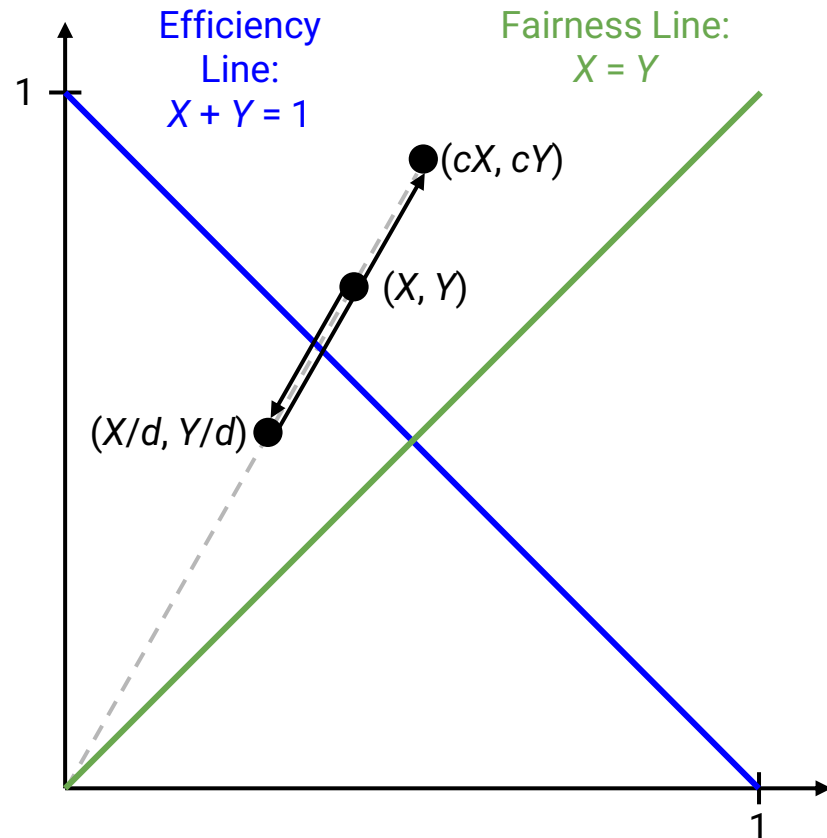
MIMD (Multiplicative Increase, Multiplicative Decrease) Adjustments on Graph

Increase: $\times c$

Decrease: $\div d$

We keep moving along the line connected to the origin.

MIMD does not converge to fairness.



MIAD (Multiplicative Increase, Additive Decrease) Dynamics

Increase: $\times 2$

Decrease: -1

$C = 5$

Start:	$X = 1$	$Y = 3$	sum = 4, no congestion	$Y - X = 2$
First iteration:	$X = 2$	$Y = 6$	sum = 8, congestion	$Y - X = 4$
Second iteration:	$X = 1$	$Y = 5$	sum = 6, congestion	$Y - X = 4$
Third iteration:	$X = 0$	$Y = 4$	sum = 4, no congestion	$Y - X = 4$
Fourth iteration:	$X = 0$	$Y = 8$	X stuck at 0 forever!	$Y - X = 8$

The gap in allocation doubles when we increase. before: $Y - X$. after: $2Y - 2X = 2(Y - X)$

The gap in allocation stays the same when we decrease. (Same as AIAD.)

MIAD is *maximally unfair*. The gap increases until one person has all the bandwidth!

AIMD (Additive Increase, Multiplicative Decrease) Dynamics

Increase: +1		Decrease: $\div 2$		$C = 5$	
Start:	$X = 1$	$Y = 2$	sum = 3, ok	$Y - X = 1$	
First iteration:	$X = 2$	$Y = 3$	sum = 5, ok	$Y - X = 1$	
Second iteration:	$X = 3$	$Y = 4$	sum = 7, congestion	$Y - X = 1$	
Third iteration:	$X = 1.5$	$Y = 2$	sum = 3.5, ok	$Y - X = 0.5$	
Fourth iteration:	$X = 2.5$	$Y = 3$	sum = 5.5, congestion	$Y - X = 0.5$	
Fifth iteration:	$X = 1.25$	$Y = 1.5$	sum = 2.75, ok	$Y - X = 0.25$	
Sixth iteration:	$X = 2.25$	$Y = 2.5$	sum = 4.75, ok	$Y - X = 0.25$	
Seventh iteration:	$X = 3.25$	$Y = 3.5$	sum = 6.75, congestion	$Y - X = 0.25$	
Eighth iteration:	$X = 1.625$	$Y = 1.75$	sum = 3.375, ok	$Y - X = 0.125$	
Ninth iteration:	$X = 2.625$	$Y = 2.75$	approaching 2.5 each!	$Y - X = 0.125$	

AIMD (Additive Increase, Multiplicative Decrease) Dynamics

The gap in allocation stays the same when we increase.

The gap in allocation halves when we decrease.

- Before: $Y - X$
- After: $(0.5 Y - 0.5 X) = 0.5(Y - X)$

The gap approaches 0, so AIMD approaches fairness!

$$Y - X = 1$$

$$Y - X = 1$$

$$Y - X = 1$$

$$Y - X = 0.5$$

$$Y - X = 0.5$$

$$Y - X = 0.25$$

$$Y - X = 0.25$$

$$Y - X = 0.25$$

$$Y - X = 0.125$$

$$Y - X = 0.125$$

AIMD (Additive Increase, Multiplicative Decrease) Adjustments on Graph

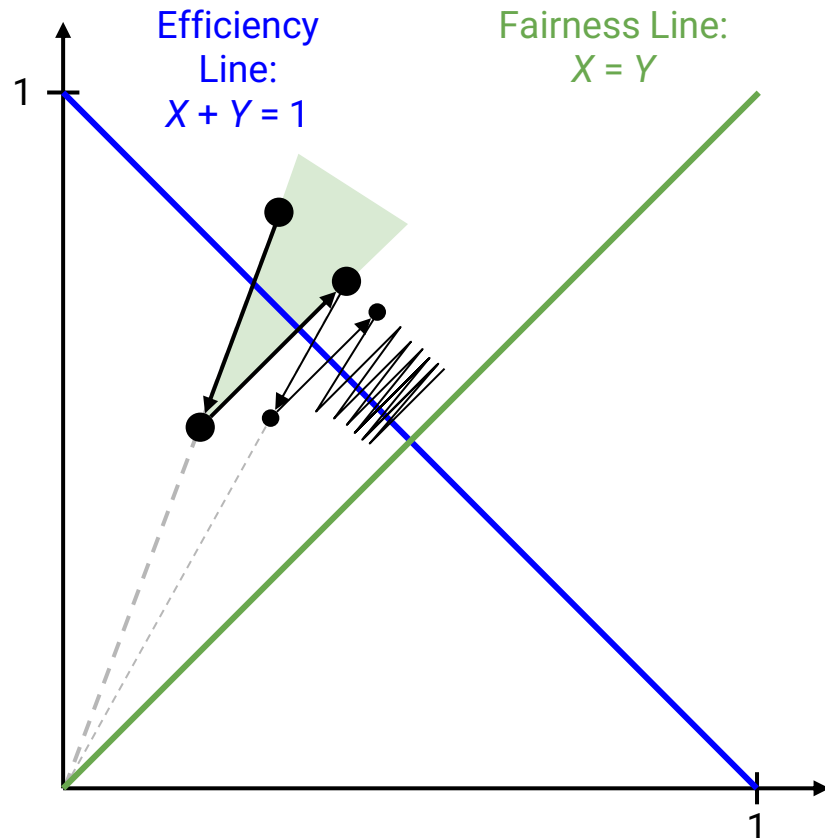
Increase: $+a$

Decrease: $\div d$

AIMD (slow-start/gentle increase, rapid decrease) converges to fairness!

Of the four options:

- AIAD and MIMD retain unfairness.
- MIAD approaches maximum unfairness (one person with all the bandwidth).
- AIMD is the only one that ensures fairness.

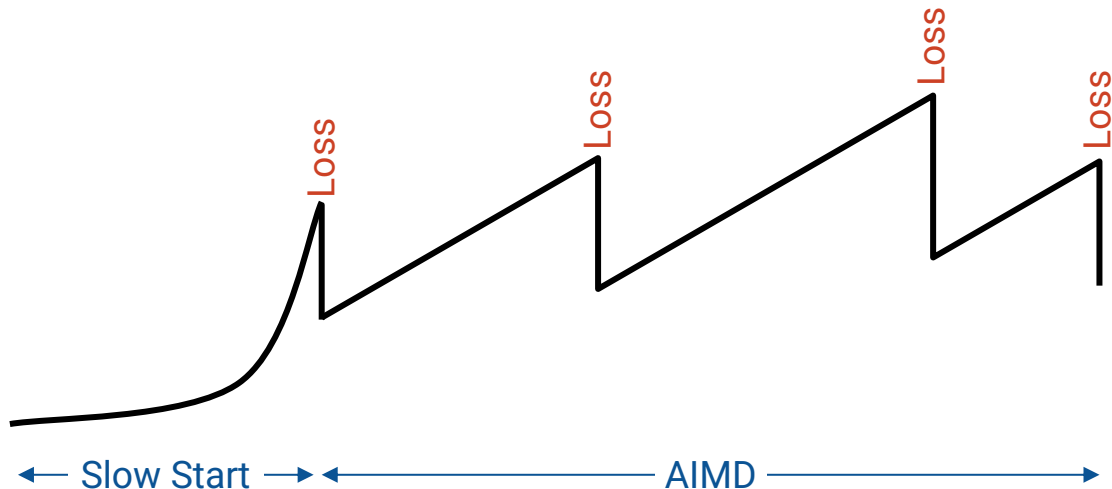


Host-Based Dynamic Adjustment: Algorithm Sketch

Every host *independently* runs the same algorithm:

1. Pick an initial rate R . (*Slow start.*)
2. Try sending at rate R for some period of time.
 - Did I experience congestion in this time period? (*Detect packet loss.*)
 - If no, increase R . (*Additive increase.*)
 - If yes, reduce R . (*Multiplicative decrease.*)
 - Repeat.

This algorithm creates a sawtooth-shaped graph as the rate is adjusted.



Implementing Congestion Control with Event-Driven Updates

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The sender maintains a window of W packets in flight.

Pick window size W to balance three goals:

1. Take advantage of network capacity ("fill the pipe").
 - Not actually used (impractical to calculate, and never the minimum).
2. Don't overload the recipient.
 - Receiver window (RWND): Recipient reports how much space it has left.
3. Don't overload links.
 - Congestion window (CWND): Sender runs a congestion control algorithm.

W is the minimum of RWND and CWND.

- For today, we'll assume $RWND > CWND$, so window is determined by CWND.
- Often true in practice too.

Real TCP measures data in bytes.

- Thus, real TCP maintains CWND in bytes.

In this lecture, we'll measure CWND in packets for convenience.

- To convert to bytes, recall: 1 packet = MSS bytes.
- MSS = Maximum Segment Size.

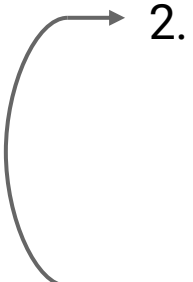
Congestion control is used to set the window size (CWND).

- Recall: $\text{Rate} = \text{CWND} / \text{RTT}$.
- Larger window = higher rate.

Every host *independently* runs the same algorithm:

1. Pick an initial rate R . (*Slow start.*)

2. Try sending at rate R for some period of time. ???

- Did I experience congestion in this time period? (*Detect packet loss.*)
 - If no, increase R . (*Additive increase.*)
 - If yes, reduce R . (*Multiplicative decrease.*)
 - Repeat.
- 

How long is "some period of time"?

- Ideally, each iteration lasts for one RTT.
- Problem: RTT is hard to measure, and changes dynamically.

Instead of updating after "some period of time," CWND updates are **event-driven**.

Sender adjusts CWND each time something interesting happens in TCP.

- We get an ack for new data.
 - This means there was no congestion.
 - Increase CWND (additive increase in AIMD).
- We get 3 duplicate acks and declare a packet lost.
 - Probably an isolated loss, since we're still getting subsequent acks.
 - Decrease CWND (multiplicative decrease in AIMD).
- The timer expires and we declare a packet lost.
 - We probably lost several packets – didn't get any duplicate acks. Bad news!
 - Abandon current rate and start over from slow-start.

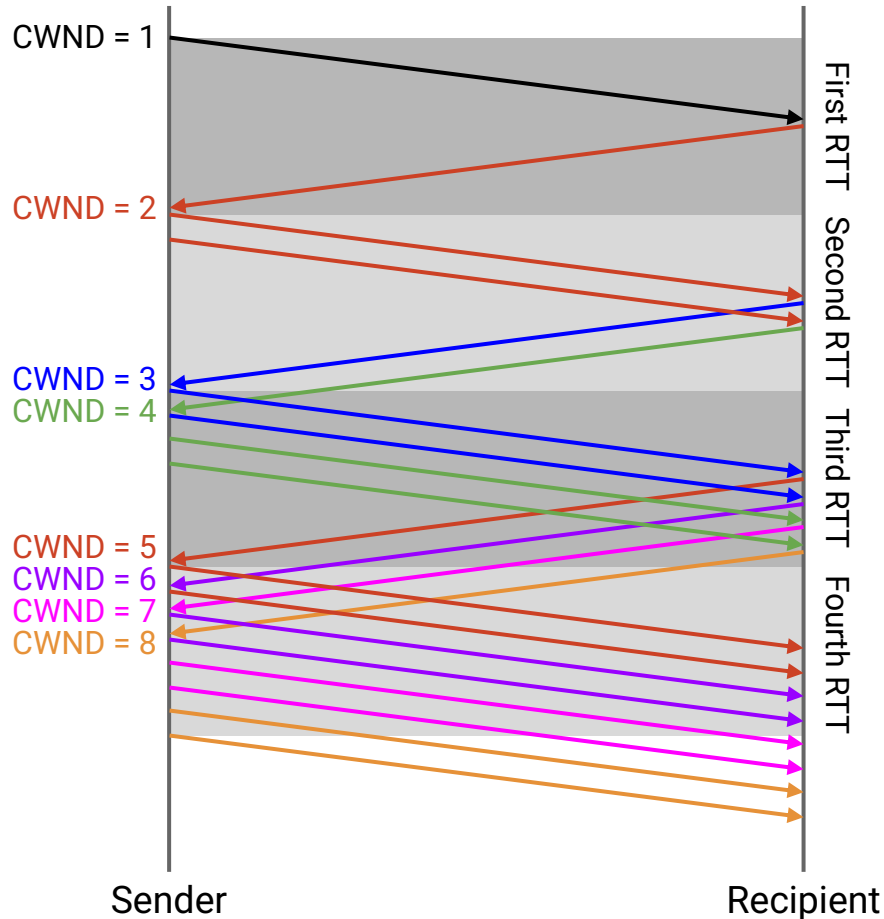
Conceptual slow start: Double CWND after each RTT.

Event-driven slow start: Increase CWND by 1 after each ack.

- Each time we get an ack, we can send one more packet (sliding window).
- But, we can also increase CWND by 1 and send another packet.
- In total: Each time we get an ack, we can send out 2 packets.
- Example: In one RTT, if we get 8 acks, we can send out 16 packets.

t = 0 RTTs:		CWND = 1	Send 1 packet
t = 1 RTT:	get 1 ack	CWND = 2	Send 2 packets
t = 2 RTTs:	get 2 acks	CWND = 4	Send 4 packets
t = 3 RTTs:	get 4 acks	CWND = 8	Send 8 packets
t = 4 RTTs:	get 8 acks	CWND = 16	Send 16 packets

Event-Driven Slow Start



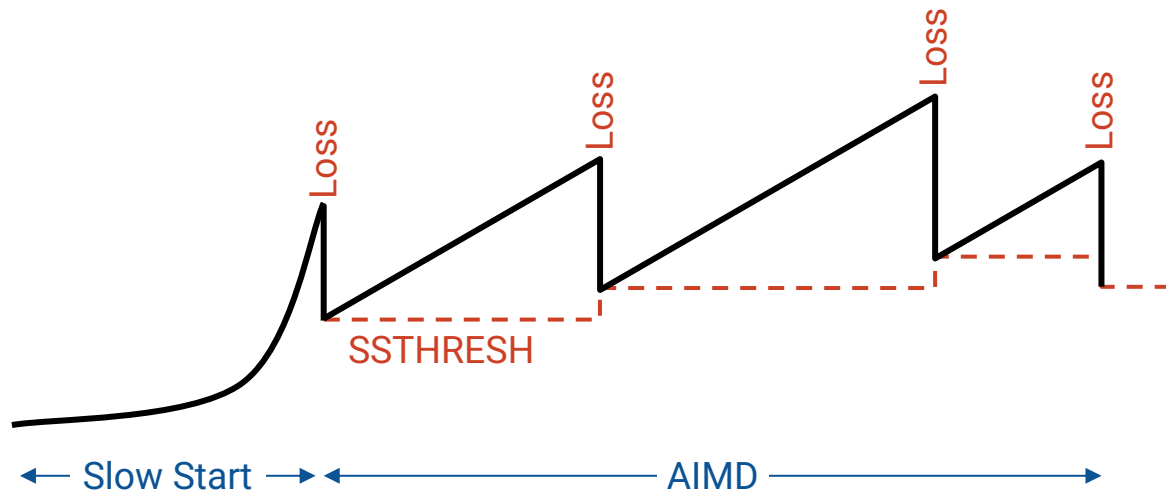
Event-Driven Slow Start: SSTHRESH

Increase CWND by 1 after each ack.

Eventually, we encounter a packet loss.

- Set $\text{CWND} = \text{CWND}/2$.
- Set $\text{SSTHRESH} = \text{CWND}/2$.

SSTHRESH helps us remember the last safe rate.



Conceptual AIMD increase: Add 1 to CWND after each RTT.

Event-driven AIMD increase: Increase CWND by $1/\text{CWND}$ after each ack.

Intuition:

- Increase CWND by a fraction of a packet on each ack.
- After one RTT's worth of packets, CWND will increase by 1.

Implementation:

- $$\text{CWND} \leftarrow \text{CWND} + \frac{1}{\text{CWND}}$$

Note: Technically, increase on each ack is different, but approximation is close enough.

- In bytes:

$$\text{CWND} \leftarrow \text{CWND} + \frac{\text{MSS}}{\text{CWND}} \times \text{MSS}$$

($1/\text{CWND}$ in packets) fraction rewritten in bytes by multiplying top and bottom by packet size: (MSS/CWND in bytes)

Total increase after each RTT should be MSS bytes, not 1 packet.

First update: $3.00 + (1/3.00) = 3.33$

Second update: $3.33 + (1/3.33) = 3.63$

Third update: $3.63 + (1/3.63) = 3.93 \approx 4$

Event-Driven AIMD Adjustments: Additive Increase

Event-driven AIMD increase: Increase CWND by a fraction after each ack.

After one RTT (aka one window) of packets, CWND increases by 1.

	Measured in packets:	Measured in bytes: ($MSS = 50$ bytes)
Old CWND:	3 packets	150 bytes (3 packets)
Fraction of increase per ack:	$1/CWND = 1/3$	$MSS/CWND = 50/150$
Total increase after 1 RTT:	+1 packet	$MSS = +50$ bytes
Increase after each ack:	+1 packet $\times 1/3 = +1/3$	+50 bytes $\times 50/150 = +50/3$
Increase after 3 acks:	$3 + (1/3) + (1/3) + (1/3)$ = 4 packets	$150 + (50/3) + (50/3) + (50/3)$ = 200 bytes

Conceptual AIMD decrease: Divide $CWND$ in half when loss detected.

Event-driven AIMD decrease: Divide $CWND$ in half when 3 *duplicate* acks detected.

- Set $CWND \leftarrow CWND / 2$
- Set $SSTHRESH \leftarrow CWND / 2$. (*Remember the last safe rate.*)

$CWND$ should never decrease below 1 packet (MSS bytes).

- If dividing by 2 causes $CWND$ to fall below 1, it's okay to round up to 1 packet.

When a packet is lost from timeout, abandon current rate and restart from slow-start.

- Rationale: We didn't get any duplicate acks before timing out, so we probably lost multiple packets.
- The current *CWND* might be way off. Safest to rediscover a good rate from scratch.
- This design decision errs on the side of caution.

Implementation: When timeout occurs:

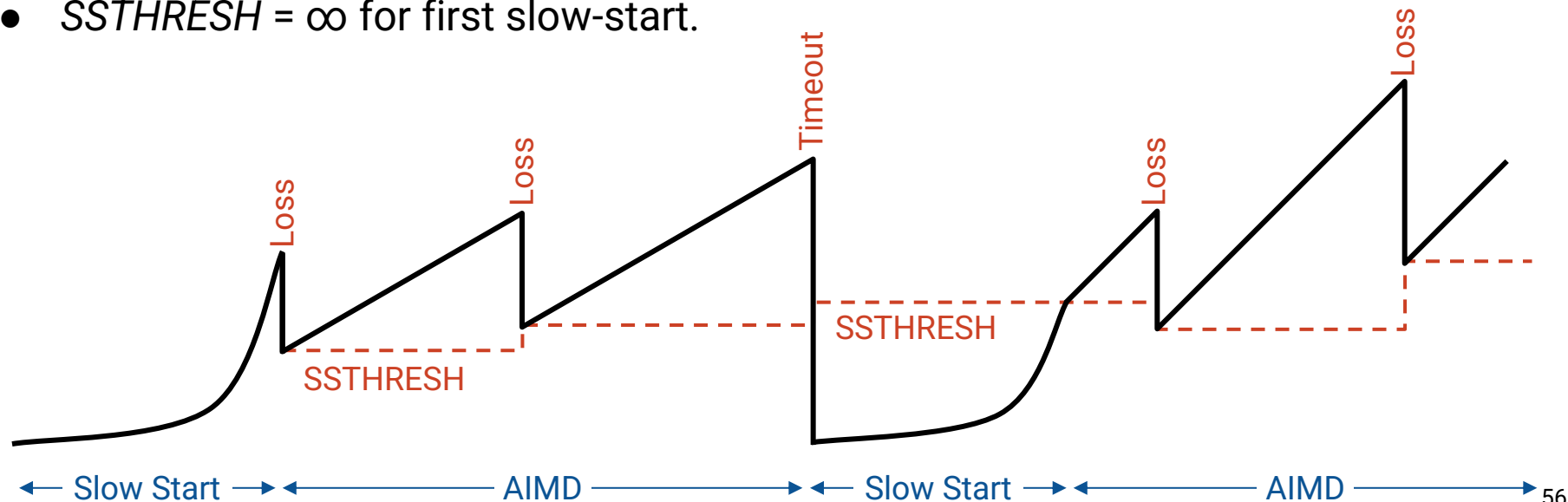
- Set $SSTHRESH \leftarrow CWND/2$. (*Remember the last safe rate.*)
- Set $CWND \leftarrow 1$ packet, and re-enter slow start mode.

SSTHRESH helps us remember the last safe rate.

- When slow-start detects loss: Update $SSTHRESH \leftarrow CWND/2$.
- When a packet times out in AIMD: Update $SSTHRESH \leftarrow CWND/2$.

During slow start: If $CWND$ exceeds $SSTHRESH$, switch to additive increases.

- $SSTHRESH = \infty$ for first slow-start.



Detecting congestion:

- Loss-based. (Delay-based is the alternative.)

Discovering an initial rate:

- Slow-start. Double rate until loss or safe rate (*SSTHRESH*) exceeded.
- Event-driven: Add 1 to *CWND* on each ack.

Adapting rate to congestion:

- AIMD (additive increase, multiplicative decrease) approaches fairness.
- Event-driven:
 - Add $1/CWND$ on each ack. After one RTT, *CWND* will have increased by 1.
 - Divide *CWND* in half when there are 3 duplicate acks.
- When a packet is lost on timeout, restart from slow-start.